



UNIVERSITY OF BLIDA1 FACULTY OF SCIENCES
COMPUTER SIENCE DEPARTMENT LMD L1 2025/2026

ALGEBRA I

Worksheet n^o : 1

Logic & Methods of proof

Exercise (01): Let P,Q and R be propositions. Give the truth table of the following compound propositions. Say which of them is true:

1. $(P \Rightarrow Q) \vee (Q \Rightarrow P)$.
2. $(P \Rightarrow (Q \wedge R)) \Leftrightarrow ((P \Rightarrow Q) \wedge (P \Rightarrow R))$.
3. $(\overline{P} \vee \overline{Q}) \Rightarrow (\overline{P \wedge Q})$.
4. $(\overline{P} \vee Q) \Rightarrow (Q \wedge (P \Leftrightarrow Q))$.
5. $((P \wedge Q) \Rightarrow (\overline{P} \vee Q))$.

Exercise (02): Using logical connectors and quantifiers, translate the following propositions and then give their negations:

1. For all $a \in]0, +\infty[$ the equation $x^2 + ax + 1 = 0$ has at least one real solution.
2. There exists a real number β such that for all real x , we have $\beta + 2\beta x = 4x + 2$.
3. For every real number a , if a satisfies $a^2 - a - 6 \leq 0$, then $a \leq -2$ or $a \geq 3$.
4. For $n \in \mathbb{N}$, if $3n^2 + n + 2$ is odd then n is odd.
5. For any positive integer n , $n^2 + 1$ is not a perfect square integer.
6. Some real numbers are greater than their squares.
7. $(U_n)_{n \in \mathbb{N}}$ is a decreasing sequence from his fifth term.
8. The graphs of the functions f and g intersect at a single point in the interval $]0, +\infty[$.
9. The function f doesn't vanish on $\mathbb{R}_+$.
10. f is not a constant function on $\mathbb{R}$.
11. f is a bounded function in $\mathbb{R}$.
12. f is a function which his values are in $[-1, 1]$. Give an example.

Exercise (03): Study the truth of the following propositions:

1. $\forall y \in \mathbb{N}, \exists x \in \mathbb{N}, y = x + 1$.
2. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, 2x - y = 5$.
3. $\forall y \in \mathbb{R}, \forall x \in \mathbb{R}, y + x > 0$.
4. $(\exists x \in \mathbb{R}, x + 1 = 0) \wedge (\exists x \in \mathbb{R}, x + 2 = 0)$.
5. $\forall n \in \mathbb{N}, (4 \text{ divides } n^2) \vee (4 \text{ divides } n^2 - 1)$.

Exercise (04):

1. Show that $(\exists x \in \mathbb{R}, x^2 = -1) \Rightarrow (\exists x \in \mathbb{R}, x^4 - 2x^2 = 3)$.
2. Proof that $\forall x, y \in \mathbb{R} - \{\frac{3}{2}\}, 4xy - 6x - 6y + 12 \neq 3$.
3. $\sqrt{2}, \frac{\ln(2)}{\ln(3)}, \sqrt{\frac{n}{n+1}}, \forall n \in \mathbb{N}^*$ are irrational numbers.
4. Let $a, b \in \mathbb{Q}$. Proof that

$$a + b\sqrt{2} = 0 \iff (a = 0 \wedge b = 0).$$
5. Proof that $\forall a, b \in \mathbb{R}$, if $a + b$ is irrational then a or b are irrational. Is the converse of this proposition always true?
6. $\forall a, b \in \mathbb{R}, ((\forall \varepsilon > 0, |a - b| \leq \varepsilon) \Rightarrow a = b)$.
7. Proof that $\forall n \in \mathbb{N} - \{0, 1, 2, 3\}$, we have $n^2 \leq 2^n$.¹
8. Proof that $\forall x \in \mathbb{R}, (x > 1 \Rightarrow E(\frac{1}{x}) = 0)$.²
9. Proof that $\forall n \in \mathbb{N}, ((n^2 - 1 \text{ is not divisible by } 8) \Rightarrow n \text{ is an even number})$.
10. Proof that $\forall n \in \mathbb{N}, \sum_{k=0}^n k k! = (n + 1)! - 1$.
11. Proof that if $(a_n)_{n \in \mathbb{N}}$ is a recurrent sequence such that $a_0 = 0$ and $a_{n+1} = a_0 + a_1 + a_2 + \dots + a_n + (n + 1)$ then $a_n = 2^n - 1$ for all $n \in \mathbb{N}$.
12. Proof that, for all $x \in \mathbb{R}, |x + 2| \leq x^2 + 5x + 7$.
13. Proof that, for all $x, y \in \mathbb{R}, x^2 + y^2 = 1 \implies |x + y| \leq \sqrt{2}$.
14. Proof that, for all $n \in \mathbb{N}^*, \sqrt{n^2 + 1} \notin \mathbb{N}$.
15. Show that the equation $3 \cos(x) = 7x^2 + 2x + 5$ has no solution for any $x \in \mathbb{R}$.
16. Let $x, y \in \mathbb{R}$, proof that

$$(|x| \leq 1 \wedge |y| \leq 1) \implies |x + y| + |x - y| \leq 2.$$

¹Use the fact that $2 < n \Rightarrow 2n < n^2 \Rightarrow 2n \leq n^2 - 1$.

² $E(x)$ is the integer part of x